MD-03 and MD-04: Proofs by Reference

Komlós and Beck–Fiala discrepancy

George Stepaniants

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, California, USA

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Source and logical status

The problems are the matrix statements in [MD03, MD04]. The cited mathematical input is Guo, Fang, and Lu, *Vector Balancing via Directional Total Variation* [GFL26], specifically **Theorem 1.1, p. 1**, and **Corollary 1.2, p. 2**. Both give the requested estimates with

$$\boxed{C_* = 3\sqrt{2\pi}.}$$

Qualification. Reference [GFL26] is the preprint version submitted on 10 September 2026. The conclusions below are conditional on the correctness of its cited results. This document supplies the complete applications of those results to the two problem statements; it is not an independent verification of the preprint's proof or a claim of a refereed resolution. The version and theorem locations were checked on 11 September 2026. A separate Codex agent independently reviewed this application note and its cited results on 11 September 2026. This review is not human peer review or a formal verification certificate.

For a real matrix $A \in \mathbb{R}^{m\times n}$, write

$$\operatorname{disc}(A) := \min_{\varepsilon\in\{-1,1\}^n} \|A\varepsilon\|_\infty, \qquad \|y\|_\infty := \max_{1\le i\le m} |y_i|.$$

The minimum is attained because its domain is a finite, nonempty set.

1 MD-03: the Komlós discrepancy statement

Theorem 1 (MD-03, using [GFL26, Theorem 1.1])**.** *Let $m, n \ge 1$ be integers and let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ satisfy*

$$\sum_{i=1}^{m} a_{ij}^2 \le 1 \qquad (1 \le j \le n). \tag{1}$$

Then, subject to the source qualification above, there exists $\varepsilon \in \{-1,1\}^n$ such that

$$\|A\varepsilon\|_\infty < C_*. \tag{2}$$

In particular, the constant requested in MD-03 can be taken to be $C = C_*$, independently of m, n and A.

Proof by citation. Let $v_j = A_{\cdot j} \in \mathbb{R}^m$. Condition (1) is exactly $\|v_j\|_2 \le 1$. Apply [GFL26, Theorem 1.1, p. 1] to these n vectors. Its signed sum is $\sum_j \varepsilon_j v_j = A\varepsilon$, so its conclusion is (2). The strict estimate implies the non-strict estimate asked for in [MD03]. $\square$

2 MD-04: the Beck–Fiala discrepancy statement

Theorem 2 (MD-04, using [GFL26, Corollary 1.2])**.** *Let $m, n \ge 1$ be integers, let t be an integer with $1 \le t \le m$, and let $A = (a_{ij}) \in \{0,1\}^{m\times n}$ satisfy*

$$\sum_{i=1}^{m} a_{ij} \le t \qquad (1 \le j \le n). \tag{3}$$

Then, subject to the source qualification above, there exists $\varepsilon \in \{-1,1\}^n$ such that

$$\|A\varepsilon\|_\infty < C_* \sqrt{t}. \tag{4}$$

Thus $C = C_*$ also meets the requirement of MD-04, independently of m, n, t and A.

Proof by citation. For a 0–1 column, its sum equals its number of nonzero entries. Consequently (3) is precisely the sparsity hypothesis of [GFL26, Corollary 1.2, p. 2]. That result gives $\operatorname{disc}(A) < 3\sqrt{2\pi t} = C_*\sqrt{t}$. A signing attaining the defining minimum proves (4), and hence the non-strict inequality in [MD04]. $\square$

Attribution and scope. George Stepaniants is the author of this application note. Both estimates are results asserted and proved in [GFL26]; neither is a new result of this note. The Beck–Fiala estimate is already a stated corollary there, not a new corollary claimed here. The signings in the two statements assign a sign to every column. The problems [MD03, MD04] ask for existence, so no algorithmic assertion is needed.

References

[GFL26] Shengtao Guo, Ethan X. Fang, and Junwei Lu. *Vector Balancing via Directional Total Variation.* arXiv:2609.11189v1 [math.CO], 10 September 2026. Theorem 1.1, p. 1 (Komlós); Corollary 1.2, p. 2 (Beck–Fiala). Versioned arXiv record; versioned PDF. Accessed 11 September 2026.

[MD03] `MColbrook/OpenProblemsInNLA`. *MD-03 — The Komlós discrepancy conjecture.* Problem statement, `matrix-discrepancy-and-optimization/ MD-03/README.md`. GitHub source. Accessed 11 September 2026.

[MD04] `MColbrook/OpenProblemsInNLA`. *MD-04 — The Beck–Fiala discrepancy conjecture.* Problem statement, `matrix-discrepancy-and-optimization/ MD-04/README.md`. GitHub source. Accessed 11 September 2026.